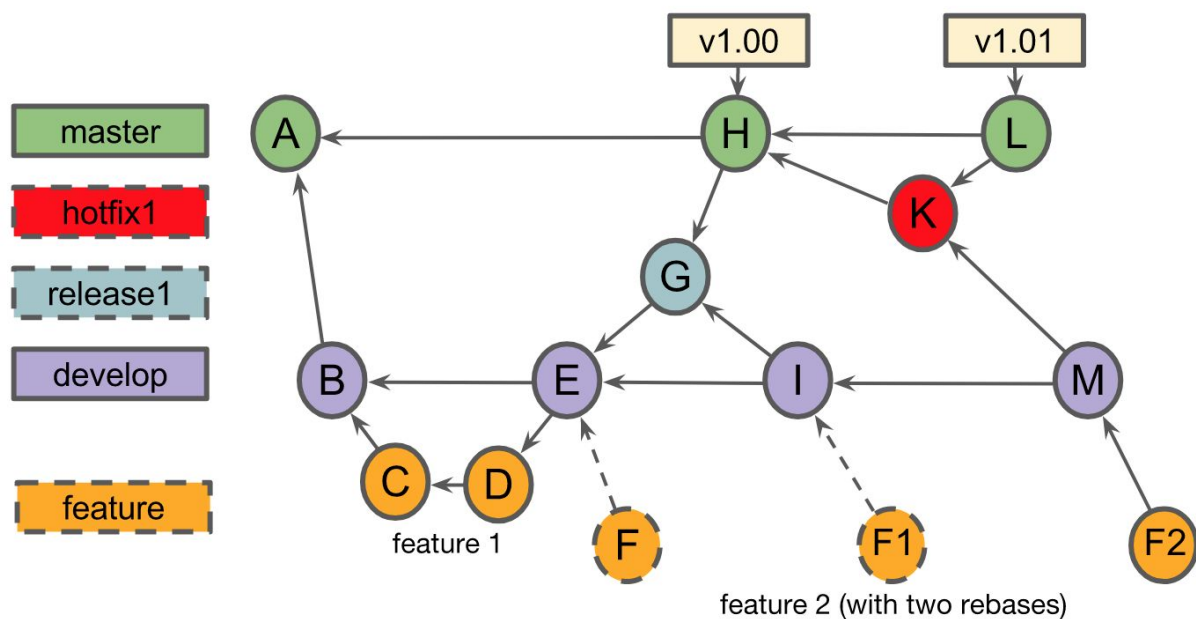


## Version Control with Git

### final project

Create a Git repository with the commits shown in the commit graph and table below. This simulates a team building and releasing a product using the Gitflow workflow. The first release (v1.00) has only one feature. Soon after release, a bug was discovered and a hotfix (v1.01) was necessary.

When you are finished, find someone to check your work. They can clone your repository to check it. Feel free to experiment and good luck!



| commit | type         | fileA.txt content       | create on branch | commit message                       | parents | notes                 |
|--------|--------------|-------------------------|------------------|--------------------------------------|---------|-----------------------|
| A      |              | -                       | master           | add README.md                        | -       | create README.md      |
| B      |              | "" (empty string)       | develop          | add fileA.txt                        | A       |                       |
| C      |              | "feature 1 wip"         | feature1         | feature 1 wip                        | B       |                       |
| D      |              | "feature 1 with 2 bugs" | feature1         | add feature 1                        | C       |                       |
| E      | merge commit | (no change)             | develop          | Merge branch 'feature1' into develop | B, D    | delete feature1 label |

|    |                 |                                          |          |                                            |      |                                           |
|----|-----------------|------------------------------------------|----------|--------------------------------------------|------|-------------------------------------------|
| F  |                 | "feature 1 with 2 bugs<br>feature 2 wip" | feature2 | feature 2 wip                              | E    |                                           |
| G  |                 | "feature 1 with 1 bug"                   | release1 | fix feature 1 bug X                        | E    |                                           |
| H  | merge<br>commit | results in<br>"feature 1 with 1 bug"     | master   | Merge branch<br>'release1'                 | A, G | tag "v1.00"                               |
| I  | merge<br>commit | results in<br>"feature 1 with 1 bug"     | develop  | Merge branch<br>'release1' into<br>develop | E, G | delete release1<br>label                  |
| F1 | rebase          | "feature 1 with 1 bug<br>feature 2 wip"  | feature2 | feature 2 wip                              | I    | will need to<br>resolve merge<br>conflict |
| K  |                 | "feature 1"                              | hotfix1  | fix feature 1 bug Y                        | H    |                                           |
| L  | merge<br>commit | "feature 1"                              | master   | Merge branch<br>'hotfix1'                  | H, K | tag "v1.01"                               |
| M  | merge<br>commit | "feature 1"                              | develop  | Merge branch<br>'hotfix1' into develop     | I, K | delete hotfix1<br>label                   |
| F2 | rebase          | "feature 1<br>feature 2 wip"             | feature2 | feature 2 wip                              | M    | will need to<br>resolve merge<br>conflict |